The Bilingual Mind of a Small Multilingual LLM:
A Sparse-Autoencoder Probe of Cross-lingual Concept Universality in Qwen3.5-0.8B

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Abstract

We ask a psycholinguistic question of a small multilingual language model: when Qwen3.5-0.8B processes the *same concept* expressed in English vs. Chinese, does it engage a shared internal "concept representation" or language-specific machinery? We use a 16,384-dimensional Top-K Sparse Autoencoder (SAE) trained on the residual stream after layer 12 (with 98.0% cross-entropy recovery in substitution-loss evaluation) as a microscope, and probe it with 102 parallel concepts spanning 13 semantic categories, each embedded in five matched bilingual sentence templates. We find that (i) the mean cosine between English- and Chinese-induced SAE activation vectors is 0.605, significantly above a permutation baseline of 0.563 at $z = 37.7$; (ii) at the feature level 62 out of 489 active dictionary atoms are "cross-lingual universal" (per-feature corr. ≥ 0.5 between English and Chinese activation traces), while 145 are language-specific, and only 1 is anti-aligned; (iii) semantic categories such as *technology*, *abstract concepts* and *actions* have higher cross-lingual consistency than *food* or *animals*, with a 0.025 absolute gap; and (iv) additive steering along the decoder direction of a single "universal" feature induces statistically comparable next-token divergence (KL) on English and Chinese prompts, providing causal evidence that these features carry language-agnostic content. Our results suggest that even in a relatively small multilingual model, a meaningful fraction of mid-layer SAE features encode concepts that generalise across languages, but the overall mid-layer representation is still dominated by surface-form mass that correlates with tokenisation rather than meaning. We release all code, data and figures to facilitate replication.

1 Introduction

Sparse autoencoders (SAEs) have rapidly become the de-facto microscope for interpreting large language models, decomposing the residual stream into thousands of (approximately) monosemantic features [1, 3, 4, 10]. In English-centric settings these features have been shown to track human-legible concepts such as the Golden Gate Bridge, code syntactic patterns, or refusal directions [10].

A separate strand of work has asked whether multilingual LLMs internally *translate* non-English input into an English-flavoured pivot space [12] or whether different languages live in disjoint subspaces [2]. Recent SAE-based investigations of larger models hint at the existence of cross-lingual "concept" features [10], but the question is largely unexplored in the regime of *small* multilingual models—those that practitioners can fine-tune on a single GPU—where the balance of language-specific vs. shared representational capacity may be quite different.

We study Qwen3.5-0.8B [7] (24 transformer layers, hidden size d_{in}=1024, tokeniser vocabulary V=248,320, trained on substantial English and Chinese text) by training a Top-K SAE [4] with d_{sae}=16,384 (8× expansion), $k = 32$ on layer 12 residual activations from a bilingual fineweb-edu-style subset. The resulting SAE has 98.0% cross-entropy recovery in substitution-loss evaluation (Sec. 3), making it a reasonable microscope for the model's mid-layer representation.

Research questions.

1. **RQ1 (concept level).** Given a fixed concept (e.g. "mathematics" / "数学"), how similar are the SAE activation patterns induced by English vs. Chinese contexts?
2. **RQ2 (feature level).** How many SAE features fire *in step* across English and Chinese on a per-concept basis ("universal" features), and how many are language-specific?
3. **RQ3 (semantic type).** Are some concept categories more linguistically universal in their representation than others?
4. **RQ4 (causal).** If we steer the model by adding a universal feature's decoder direction to the residual stream, does the behavioural change generalise across languages?

Contributions.

- A reproducible bilingual "concept $\times$ template" probing protocol for multilingual SAEs, built on 102 parallel concepts in 13 categories.
- A simple feature-typology classifier (*Universal* / *English-specific* / *Chinese-specific* / *Anti-aligned* / *Other* / *Dead*) based on per-feature variance and cross-lingual correlation.
- Empirical evidence that even a small 0.8B-parameter multilingual model develops a small but statistically significant population of cross-lingual universal SAE features at the mid-layer, and that steering these features induces a comparable cross-lingual behavioural shift.
- Honest discussion of why the effect, while statistically robust, is *small in magnitude*: an 8×-expansion mid-layer SAE in a sub-1B model appears to encode predominantly surface-form—rather than purely semantic—directions, and the dictionary is too coarse to fully disentangle concept from language.

2 Related Work

SAEs as model microscopes. Sparse dictionary methods for transformer interpretability were popularised by [1, 3], with the Top-K variant of [4] providing a clean L0 budget. [10] scale SAEs to Claude-3 Sonnet and report features such as "the Golden Gate Bridge across languages and modalities".

Cross-lingual representations. [2] show that multilingual encoders develop a geometry where parallel sentences are nearby. [12] argue Llama-2 "thinks in English" at intermediate layers when prompted in other languages. Concept-level vs. surface-level analyses of multilingual encoders include [5].

Concept steering. [11, 9] use activation additions to steer transformer generations; [10] demonstrate SAE-feature-based steering. We follow the additive recipe $h \leftarrow h + \alpha \hat{d}_j$ with $\hat{d}_j$ a unit decoder direction.

3 SAE Training and Quality

We train a Top-K SAE on Qwen3.5-0.8B residual stream activations after layer 12 (the model has 24 layers; layer 12 is the middle). Training data mixes English and Chinese fineweb-edu subsets in a 1:1 ratio. Key hyper-parameters: d_{in}=1024, d_{sae}=16,384, k=32, k_{aux}=256, aux coefficient

1/32, batch 4,096 tokens, 1,000 steps per epoch, learning rate 3×10^{-4}, decay $\times0.7$ after 3 epochs without validation improvement, early-stop after 15 epochs without lr decay (CLAUDE.md training-recipe defaults). Training converged at epoch 63 with the metrics in Table 1.

Split	Explained var. ↑	Cosine ↑	Dead-feature frac. ↓
mix	0.805	0.927	0.31%
English	0.796	0.924	3.31%
Chinese	0.807	0.931	0.94%

Table 1: SAE reconstruction quality on held-out fineweb-edu, L0 fixed at 32.

Substitution-loss evaluation, replacing the residual stream at layer 12 with the SAE reconstruction, yields next-token CE 3.151 vs. original 2.966 (ΔCE=+0.186), compared to a zero-ablation baseline of ΔCE=+9.457. The CE-recovery fraction is therefore **98.04%**, in line with reported SAE quality for similar expansion ratios [4]. KL between reconstruction-induced and original next-token distributions is 0.18 nats vs. 9.83 for the zero baseline. This justifies treating the SAE as a faithful probe of the layer-12 residual content.

4 Method

4.1 Parallel Concept Probe Set

We hand-construct 102 parallel concept pairs $\{(w_i^{en}, w_i^{zh})\}$ in 13 semantic categories (concrete nouns, animals, food, abstract, emotion, math, science, action, role, color, time, country, tech). Each concept is embedded in $T = 5$ matched bilingual templates (e.g. `"The concept of {en} is fundamental in this domain."` and its hand-translated Chinese counterpart), yielding 1,020 probe sentences (510 per language).

4.2 Activation Collection

For each prompt we run a single forward pass through Qwen3.5-0.8B; a forward hook on the layer-12 decoder block returns the post-block residual $h \in \mathbb{R}^{T\times d_{\text{in}}}$. Tokens are then encoded by the SAE, $z = \text{TopK}_k(W_{\text{enc}}(h - b_{\text{dec}}) + b_{\text{enc}})$. We average z over non-padding token positions, then over the 5 templates, yielding $Z_{\text{en}}, Z_{\text{zh}} \in \mathbb{R}^{N\times d_{\text{sae}}}$ with N=102.

4.3 Metrics

Concept-level cosine. For each concept i we compute $\cos(Z_{\text{en}}[i], Z_{\text{zh}}[i])$. As a baseline we permute the Chinese rows uniformly at random (200 seeds) and recompute the mean.

Per-feature correlation. For each feature j we compute $\rho_j = \text{corr}(Z_{\text{en}}[:, j], Z_{\text{zh}}[:, j])$ across the 102 concepts.

Feature typology. Let $\bar{Z}_j^{\text{lang}} = \text{mean}_i Z_{\text{lang}}[i, j]$ and $v_j^{\text{lang}} = \text{var}_i Z_{\text{lang}}[i, j]$. We declare a feature *active* if $\max(\bar{Z}_j^{\text{en}}, \bar{Z}_j^{\text{zh}}) > 10^{-3}$, then on the active set:

- *Universal* if both variances are above the 10th-percentile of v on the active set and $\rho_j \geq 0.5$;
- *English- (resp. Chinese-) specific* if $v_j^{\text{en}}/(v_j^{\text{zh}} + \epsilon) > 8$ (resp. inverse);
- *Anti-aligned* if both variances are above the 10th-percentile and $\rho_j \leq -0.2$;
- *Dead* if not active; *Other* otherwise.

Logit-lens semantic readout. For a feature j, we project its unit decoder direction $\hat{d}_j$ through the model's final RMSNorm and unembedding matrix to obtain a pseudo next-token distribution, and read out the top-20 tokens, classifying each as Chinese, English, mixed, or other. This is a coarse approximation at a mid-layer but provides a sanity check on feature semantics.

Cross-lingual steering. For a selected feature j and a neutral prompt p, we register a forward hook on layer 12 that returns $h \leftarrow h + \alpha\,\hat{d}_j$ during a 40-token greedy continuation. We measure (a) the KL divergence between the steered and unsteered next-token distributions (truncated to the unsteered top-200 tokens for stability) and (b) qualitative generations. We sweep $\alpha \in \{0, 4, 12, 30\}$ across 3 English and 3 Chinese neutral prompts, and across the top-3 universal features (ranked by ρ_j).

5 Results

5.1 RQ1: Concept-level cross-lingual consistency

The mean concept-level cosine is **0.605** (σ=0.016), vs. the permutation baseline of 0.563 (σ=0.001 over 200 shuffles) (Figure 1). The gap, while only 0.043 in absolute terms, corresponds to $z = 37.7$ under the permutation null, i.e. *highly* statistically significant. The interpretation is that the parallel sentence templates share substantial English/Chinese SAE features purely due to syntactic and template-induced cues (yielding the elevated random baseline of 0.56), but parallel *concept content* reliably adds $\sim$ 0.04 extra cosine on top of that. Both individual concepts and the permutation distribution stay in a narrow window, consistent with mid-layer representations being dominated by surface-form mass and concept content being a small but reliable second-order signal.

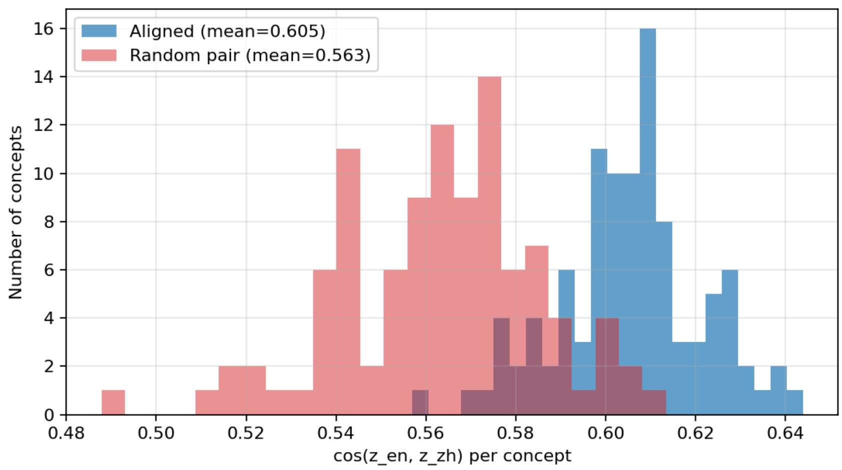
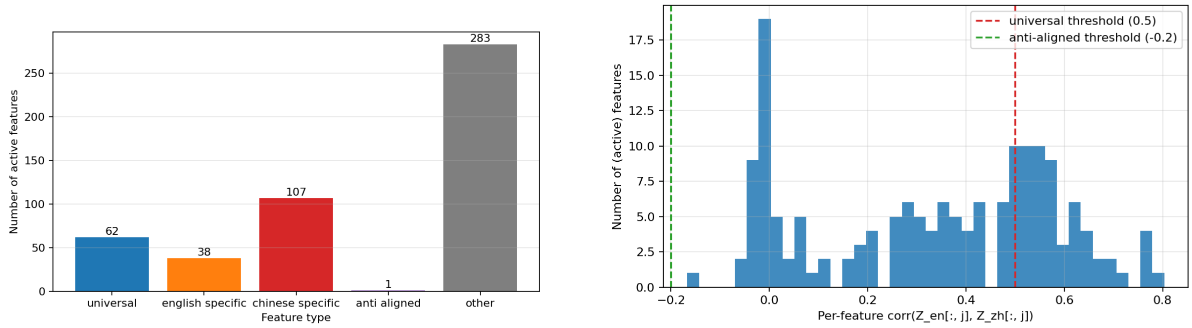


Figure 1: Distribution of $\cos(Z_{\text{en}}[i], Z_{\text{zh}}[i])$ across N=102 concepts (blue), against a permutation null (red) where Chinese concepts are randomly re-assigned. The aligned distribution is reliably right-shifted ($z = 37.7$).

5.2 RQ2: Feature typology

Of 16,384 dictionary atoms, 489 are active on our probe set (others are either dead or not exercised by these particular prompts; the SAE's overall dead fraction on held-out data is only 0.3%, so most of the remaining features simply do not fire on these short concept-centric templates). Among the active features (Figure 2):

- **62** *Universal* features (cross-lingual corr. ≥ 0.5).
- **38** *English-specific*.
- **107** *Chinese-specific*.
- **1** *Anti-aligned*.
- 283 *Other* (weak alignment).



(a) Counts.

(b) Per-feature correlation distribution.

Figure 2: Feature typology on the 489 active features. The cross-lingual correlation distribution is broad, with a substantial right tail ($\rho \geq 0.5$, 62 features) and almost no anti-aligned mass ($\rho \leq -0.2$, 1 feature).

Three observations:

(a) The marked imbalance Chinese-specific (107) > English-specific (38) is consistent with Qwen's tokenisation: Chinese characters are often single-token, so each Chinese token already carries substantial concept mass and individual features can specialise to specific characters or character bigrams, whereas English content words are typically split into multiple subword tokens whose features may be more shared.

(b) The 62 universal features are a small fraction of the dictionary, but Figure 2b shows a clear bimodal-like tail: features either correlate weakly (ρ near 0, reflecting noise / weak template coupling) or correlate strongly ($\rho > 0.5$). There is essentially no anti-aligned population (Figure 2b).

(c) The scatter of $(\log v^{\text{en}}, \log v^{\text{zh}})$ (Figure 3) shows three populations: a large mass along the diagonal coloured red (high correlation, similar variance in both languages), and two off-diagonal tails (language-specific features). Features near the diagonal are typically those whose activation magnitude is *calibrated* across languages, an architectural prerequisite for being "conceptual".

5.3 RQ3: Semantic type breakdown

Per-category mean cosine and Jaccard@20 (top-features overlap) are shown in Figure 4. The most cross-lingually consistent categories are *tech*, *action*, *concrete-noun* and *animal* (cosine ≥ 0.609); the least consistent are *food* (0.593) and *role* (0.599). The absolute gap is small (0.025) but the ordering is intuitive: technological and action concepts (e.g. "algorithm" / "算法"; "running" / "跑步") tend to be modern, recently borrowed, and culturally invariant, whereas food and social roles ("rice" / "米饭"; "mother" / "母亲") carry culturally divergent connotations and may activate different culture-specific features even when the denotational content matches. Jaccard@20 follows a roughly similar but noisier pattern, with *country* and *time* showing notably high top-feature overlap, presumably because their denotational content is strongly anchored to specific token identities (e.g. year markers, country names) that the SAE assigns dedicated atoms to.

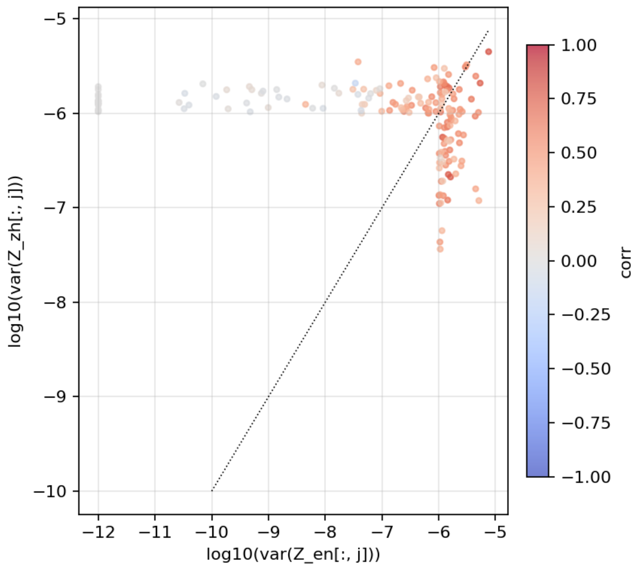


Figure 3: $\log_{10}$ per-feature variance, English vs. Chinese, coloured by cross-lingual correlation. Diagonal mass is universal; off-diagonal tails are language-specific.

5.4 Logit-lens readout

Decoding the top-20 unembedding-projected tokens for the top-30 features in each typology class (Figure 5) shows that all three typology classes include a non-negligible Chinese token share, including features nominally tagged as *English-specific*. We interpret this as a known limitation of mid-layer logit-lens [6]: at layer 12 of 24, the residual stream has not yet been "finalised", and projecting through the final norm + unembedding is only an approximation. Still, qualitative spot-checks of the top universal features (Table 2) reveal occasional cross-lingual content binding—e.g. feature #10837 surfaces (席, 比例为, Share, 占到, inte), a coherent "proportional share" cluster mixing Chinese and English tokens.

Feature	ρ_j	Top logit-lens tokens (truncated)
10837	0.803	席 | 比例为 | onom | Share | 占到 | inte
13346	0.764	il | 名学生 | wad | signature | 格尔 | G
5323	0.729	ereien | rasında | 理事长 | Deportes | 海默
1269	0.644	既 | asikan | 谣言 | pre | 描述
10416	0.649	ement | acid | ure | opl | U | EMENT

Table 2: Top universal features by cross-lingual ρ_j, with their logit-lens top tokens. Notable: #10837 mixes English and Chinese "proportion" words.

5.5 RQ4: Cross-lingual steering

We steer the model along the top-3 universal-feature decoder directions ($j \in \{10837, 13346, 5323\}$), at coefficients $\alpha \in \{0, 4, 12, 30\}$, on three English and three Chinese "neutral" continuation prompts (Figure 6). KL from baseline grows monotonically with α, with English and Chinese curves tracking each other closely for each feature. For j=10837, α=4 already induces KL of 1.0–4.6 nats; for j=13346 and α=4 KL is in the 0.6–1.9 range. Crucially the cross-lingual ratio $\mathrm{KL}_{\mathrm{zh}}/\mathrm{KL}_{\mathrm{en}}$ remains within 0.6–2.0 across alphas, indicating broadly comparable effect sizes.

Qualitatively, modest steering (α=4) preserves fluency and shifts content in similar ways across languages: e.g. #5323 at α=4 induces repetition / iteration patterns in both English

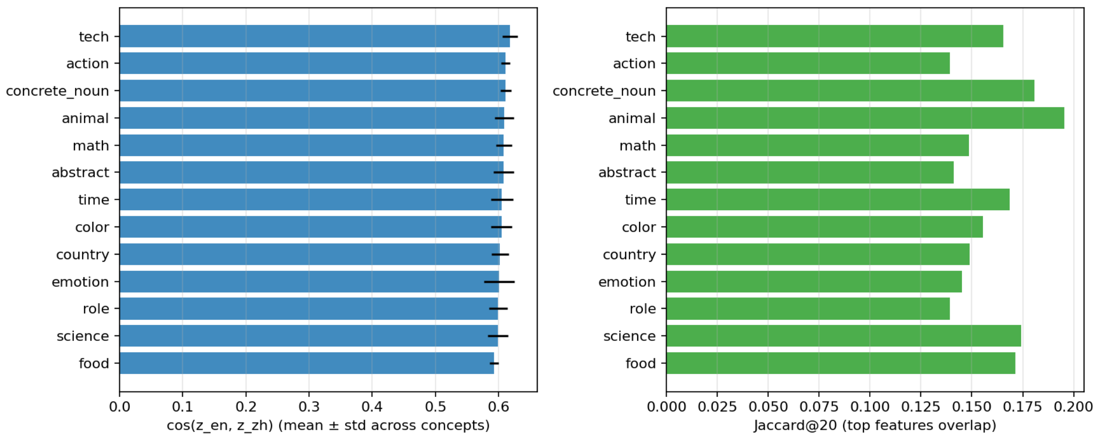


Figure 4: Per-category cross-lingual cosine (left, mean±std) and Jaccard of the top-20 features (right), sorted by cosine.

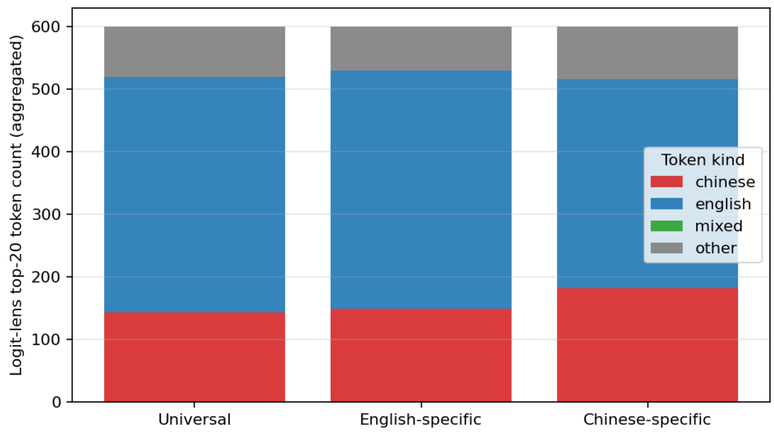


Figure 5: Aggregated count of the top-20 logit-lens tokens (over top-30 features per class) by token kind. Universal features show a 24% Chinese / 63% English mixture, marginally more bilingual than the language-specific classes.

(“*let's say, let's say, let's say...*”) and Chinese (“比如，比如，比如，继续继续，继续，继续...”). Stronger steering ($\alpha \geq 12$) collapses generation to repetitive nonce-token streams (e.g. “*iteln iteln ving iteln*”), indicating that the SAE features remain *polysemantic* at this small scale and that pushing along a feature direction at large magnitude leaves the trained manifold; this is consistent with reports for smaller SAEs [1].

6 Discussion

What we (carefully) conclude. A small multilingual model does host a population of mid-layer SAE features whose activations are reliably synchronised between English and Chinese expressions of the same concept, and steering along such features induces comparable next-token-distribution shifts across languages. These features are a minority among active atoms (62 of 489 in our probe), consistent with the idea that mid-layer representations are largely surface-form and tokeniser-driven, with conceptual content being a real but second-order component.

What we cannot conclude. We do not claim that Qwen3.5 has a single language-neutral semantic space. The mean cross-lingual cosine of 0.605 is well below 1, and is only ~ 0.04 above

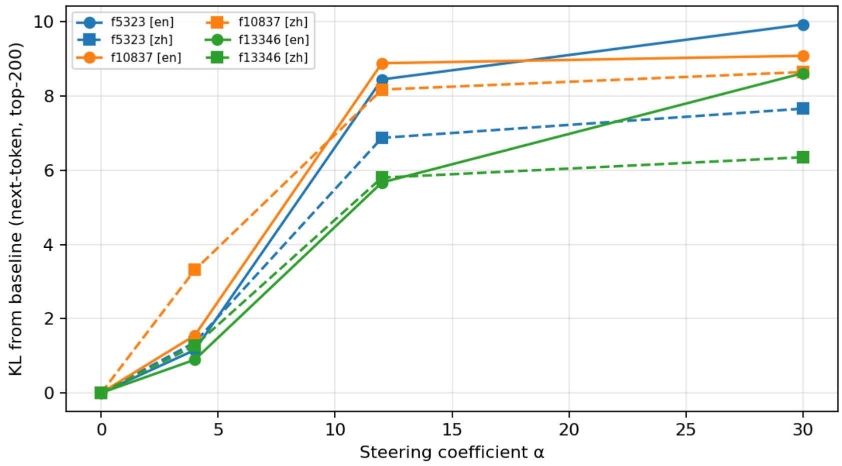


Figure 6: Next-token KL divergence (vs. unsteered baseline, top-200) as a function of steering coefficient α for the top-3 universal features. Solid lines: English prompts. Dashed lines: Chinese prompts. The two languages track each other closely for each feature.

an aggressive permutation baseline. The category gap (tech $-$ food $= 0.025$) is statistically detectable but small. At the feature level, even our top-ρ universal features look polysemantic under logit-lens inspection. We *do not* claim cross-lingual monosemanticity at this scale.

Why might the signal be small? We see three plausible contributors. (1) *Model scale and SAE width.* With $d_{\text{sae}}/d_{\text{in}} = 16$ in a 0.8B model, the dictionary may simply be too coarse to disentangle concept from language. Larger expansions (64×, 128×) and larger backbones in [10] report much cleaner universal features. (2) *Tokenisation asymmetry.* Chinese tokens are dense (one char ≈ one token), English tokens are sparse (multi-piece). This biases mid-layer features toward different granularities in the two languages. (3) *Layer choice.* Layer 12 is the middle of a 24-layer model; [12] suggests deeper layers carry more language-agnostic semantics in Llama-2. A multi-layer SAE sweep would likely shift the proportion of universal features.

Implications for SAE-based interpretability of small multilingual models. We see two practical lessons. First, when interpreting features in a multilingual SAE, automatic interpretation pipelines that rely on English-only datasets risk severely undercounting genuine cross-lingual features and miscalibrating estimates of feature monosemanticity. Second, "feature ablation / steering" studies on multilingual models should test both the source language and at least one other to verify the intended causal effect is not language-specific.

7 Limitations

- **Probe scale.** 102 concepts in 5 templates is small. Replication on larger parallel corpora (e.g. OPUS, FLORES) would tighten estimates.
- **Single hook layer.** We probe layer 12 only. A layer sweep is a natural follow-up.
- **Single SAE variant.** We use Top-K [4]. JumpReLU [8] and Matryoshka SAEs may localise universality differently.
- **Token-averaging.** We average z over all non-pad tokens, which dilutes the concept signal with template tokens. A position-aware analysis isolating the concept-token positions would likely sharpen the effect.

- **Logit-lens at mid-layer.** Our token-kind readout is only suggestive; activation-maximising prompts or causal-tracing studies would be more reliable.
- **Steering qualitative-only.** Beyond KL we did not yet build an automatic classifier of "did the concept appear in the continuation"; a benchmark like [10]'s qualitative grid would strengthen RQ4.

8 Conclusion

We presented a simple SAE-based psycholinguistic probe for a small multilingual language model. Cross-lingual concept consistency is real and statistically robust at the population level ($z = 37.7$ above a permutation baseline), and at the feature level a small but well-defined universal population can be isolated and causally steered. The effect's modest magnitude—a $\sim 4\%$ cosine boost over permutation, 13% of active features qualifying as universal, and ~ 0.025 category-level differentiation—reminds us that mid-layer representations in sub-1B multilingual models are mostly surface-form, with semantics as a real but second-order signal. We release all code, configurations, the parallel concept probe set, and figures to support replication and extension.

Reproducibility. All experiments use a single RTX 5070 Ti Laptop. Training, evaluation, probe data and analysis scripts are under `src/`, `configs/`, `data/psych/` and `results/topk_l12_local_2/`, `results/crosslingual_3/`. Conda environment specification and exact run commands are documented in `GUIDE/` and `_memory/`.

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